

A Utility for Creating Execution Environments

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Contents

- [1 Introduction](#)
- [2 Executive Summary](#)
- [3 Revision History](#)
 - [3.1 R4](#)
 - [3.2 R3](#)
 - [3.3 R2](#)
 - [3.4 R1](#)
 - [3.5 R0](#)
- [4 Discussion](#)
 - [4.1 Motivation](#)
 - [4.2 Design Considerations](#)
 - [4.2.1 Singleton Environments](#)
 - [4.2.2 Environment Composition](#)
 - [4.2.3 By Value vs. By Reference](#)
 - [4.2.4 Should Environments Be Constrained?](#)
 - [4.3 Naming](#)
- [5 Implementation Experience](#)
- [6 Future Extensions](#)
- [7 Proposed Wording](#)
- [8 Acknowledgements](#)

“The environment is everything that isn’t me.”
— Albert Einstein

§ 1 Introduction

Execution environments are a fundamental part of the sender-based facilities in `std::execution`, but the current working draft provides no utility for creating or manipulating environments. Such utilities are increasingly necessary considering that some proposed APIs (e.g., `write_env` from [P3284R0]) require passing an environment.

This paper proposes two simple utilities for creating execution environments: one to create an environment out of a query/value pair and another to join several environments into one.

§ 2 Executive Summary

This paper proposes the following changes to the working draft:

- Add a class template provisionally called `prop` to the `std::execution` namespace. `prop` associates a query `Q` with a value `V`, yielding a *queryable* object `E` such that `E.query(Q)` is equal to `V`.
- Add a class template provisionally called `env` to the `std::execution` namespace. `env` aggregates several environments into one, giving precedence to the environments in lexical order.
- Optionally, replace `empty_env` with `env<>`.

§ 3 Revision History

§ 3.1 R4

- Wording fixes requested by LWG.

§ 3.2 R3

- Delete the assignment operators of `prop` and `env`.

§ 3.3 R2

- Change the target to the working draft instead of P2300.
- For the type `prop<Query, Value>` mandate that `Query()(env)` is well-formed, where `env` is an environment whose type is equivalent to `prop<Query, Value>`.

- From the `env` class template, remove support for queries that take extra arguments.
- Attributes `[[nodiscard]]` and `[[no_unique_address]]` are removed from the specification, following LEWG’s and LWG’s policies.

§ 3.4 R1

- Added a section considering whether `prop` or `env` should enforce the syntactic requirements of the queries they support. See “[Should Environments Be Constrained?](#)”.

§ 3.5 R0

- Initial revision

§ 4 Discussion

§ 4.1 Motivation

In `[exec]`, execution environments are an internal implementation detail of the sender algorithms. There are no public-facing APIs that accept environment objects as a parameter. One may wonder why a utility for constructing environments is even necessary.

There are several reasons why the Standard Library would benefit from a utility to construct an environment:

1. The set of sender algorithms is openly extensible. For those who decide to implement their own sender algorithms, the manipulation of execution environments is part of the job. Standard utilities will make this easier.
2. [\[P3284R0\]](#) proposes a `write_env` sender adaptor that merges a user-specified environment with the environment of the receiver to which the sender is eventually connected. Using this adaptor requires the user to construct an environment. Although implementing an environment is not hard, an out-of-the-box solution will make using `write_env` simpler.
3. [\[P3149R3\]](#) proposes two new algorithms, `spawn` and `spawn_future`, both of which can be parameterized by passing an environment as an optional argument.
4. Currently, there is no way to parameterize algorithms like `sync_wait` and `start_detached`. The author intends to bring a paper proposing overloads of those functions that accept environments as a way to inject things like stop tokens, allocators, and schedulers into the asynchronous operations that those algorithms launch.

In short, environments are how users will inject dependencies into async computations. We can expect to see more APIs that will require (or will optionally allow) the user to pass

an environment. Thus, a standard utility for making environments is desirable.

§ 4.2 Design Considerations

§ 4.2.1 Singleton Environments

Defining an execution environment is quite simple. To build an environment with a single query/value pair, the following suffices:

```
template <class Query, class Value>
struct prop
{
    [[no_unique_address]] Query query_;
    Value value_;

    auto query(Query) const noexcept -> const Value &
    {
        return value_;
    }
};

// Example usage:
constexpr auto my_env = prop(get_allocator, std::allocator{});
```

Although simple, this template has some nice properties:

1. It is an aggregate so that members can be direct initialized from temporaries without so much as a move. A function that constructs and returns an environment using `prop` will benefit from RVO:

```
// The frobnicator object will be constructed directly in
// the callers stack frame:
return prop(get_frobnicator, make_a_frobnicator());
```

2. An object constructed like `prop(get_allocator, my_allocator{})` will have the simple and unsurprising type `prop<get_allocator_t, my_allocator>`.

The `prop` utility proposed in this paper is only slightly more elaborate than the `prop` class template shown above, and it shares these properties.

§ 4.2.2 Environment Composition

What if you need to construct an environment with more than one query/value pair, say, an allocator and a scheduler? A utility to join multiple environments would work together with `prop` to make a general environment-building utility. This paper calls that utility `env`.

The following code demonstrates:

```
template <class Env, class Query>
concept has-query = requires (const Env& env) { env.query(Query()); };

template <class... Envs>
```

```

struct env : Envs...
{
    template <class Query>
    static constexpr size_t _index_of() {
        constexpr bool flags[] = {has-query<Envs, Query>...};
        return ranges::find(flags, true) - flags;
    }

    template <class Query>
    requires (has-query<Envs, Query> || ...)
    decltype(auto) query(Query q) const noexcept(...)
    {
        auto tup = tie(static_cast<const Envs&>(*this)...);
        return get<_index_of<Query>()>(tup).query(q);
    }
};

template <class... Envs>
env(Envs...) -> env<Envs...>;

// Example usage:
constexpr auto my_env = env{prop(get_allocator, my_alloc{}),
                             prop(get_scheduler, my_sched{})};

```

This env class template shares the desirable properties of prop: aggregate initialization and unsprising naming. A query is handled by the “leftmost” child environment that can handle it.

Note that the above code has the issue that two child environments cannot have the same type. That is not a limitation of the facility proposed below.

§ 4.2.3 By Value vs. By Reference

There are times when you would like an environment to respond to a query with a reference to a particular object rather than a copy. Capturing a reference is dangerous, so the opt-in to reference semantics should be explicit.

std::reference_wrapper is how the Standard Library deals with such problems. It should be possible to construct an environment using std::ref to specify that the “value” should be stored by reference:

```

std::mutex mtx;
const auto my_env = prop(get_mutex, std::ref(mtx));
std::mutex & ref = get_mutex(my_env);
assert(&ref == &mtx);

```

Similarly, there are times when you would like to store the child environment by reference. std::ref should work for that case as well:

```

// At namespace scope, construct a reusable environment:
constexpr auto global_env = prop(get_frobicator, make_frobicator());

// Construct an environment with a particular value for the get_allocator

```

```
// query, and a reference to the constexpr global_env:
auto env_with_allocator(auto alloc)
{
    return env{prop(get_allocator, alloc), std::ref(global_env)};
}
```

The utility proposed in this paper satisfies all of the above use cases.

§ 4.2.4 Should Environments Be Constrained?

An interesting question came up during LEWG design review at the St. Louis committee meeting. Consider a *queryable* object *e* such that *e*.query(get_scheduler) is well-formed. The get_scheduler query should always return an object whose type satisfies the scheduler concept. So, should the expression *e*.query(get_scheduler) be somehow constrained with that requirement? Should the expression be ill-formed if it returns an int, say.

During the LEWG telecon on July 16, 2024, it was decided that instantiations of the prop<Query, Value> class template should mandate that Query is callable with prop<Query, Value> (or rather, a type equivalent to prop<Query, Value> since the prop<Query, Value> type will be incomplete at the time the requirement is checked).

This will have the effect that, if Query()(env) has requirements or mandates on the expression env.query(Query()), that those requirements or mandates will cause a hard error when they are not met, and that the error happens as early as possible: when the prop class template is instantiated.

For example, consider the following query:

```
struct get_scheduler_t
{
    template <class Env>
        requires has_query<Env, get_scheduler_t>
        decltype(auto) operator()(const Env& env) const
        {

            // Mandates: env.query(*this) returns a scheduler
            static_assert(
                scheduler<decltype(env.query(*this))>,
                "The 'get_scheduler' query must return a type that satisfies the 'scheduler' concept.");

            return env.query(*this);
        }
}

inline constexpr get_scheduler_t get_scheduler {};
```

With the code above, the expression prop(get_scheduler, 42) will cause the program to be ill-formed because the type int does not satisfy the scheduler concept.

§ 4.3 Naming

For the prop utility, a couple of other plausible names come to mind:

- attr
- with
- property
- key_value
- query_value

Other possible names for env are:

- attributes, or attrs
- properties, or props
- dictionary, or dict

§ 5 Implementation Experience

Types with a very similar designs have seen heavy use in `stdexec` for over two years. The design proposed below has been prototyped and can be found on [Compiler Explorer¹](#).

A note on implementability: The implementation of `env` is simple if we only want to support braced list initialization with class template argument deduction, like `env{e1, e2, e2}`. Initialization from a parenthesized expression list with CTAD for an arbitrary number of arguments (e.g., `env(e1, e2, e3, ...)`) is not implementable in standard C++ to the author's knowledge.

In the proposed wording, `env` is specified such that it will be able to benefit from member packs ([P3115R0]) if and when they become available or should aggregate initialization with a parenthesized expression list ever be extended to support brace elision for subobjects.

§ 6 Future Extensions

There are several ways this utility might be extended in the future:

- Add a way to *remove* a query from an environment.
- Add a wrapper for a reference to an environment.
- Add a wrapper that only accepts queries that are *forwarding* (see [\[exec.fwd.env\]](#)).

§ 7 Proposed Wording

Replace all occurrences of `empty_env` with `env<>`.

[Editor's note: Change `[exec.syn]` as follows:]

Header `<execution>` synopsis `[exec.syn]`

```
namespace std::execution {
    ...as before...

    struct empty_env {};
    struct get_env_t { see below };
    inline constexpr get_env_t get_env{};

    template<class T>
        using env_of_t = decltype(get_env(declval<T>()));

    // [exec.prop] class template prop
    template<class QueryTag, class ValueType>
    struct prop;

    // [exec.env] class template env
    template<queryable... Envs>
    struct env;

    // [exec.domain.default], execution domains
    struct default_domain;

    // [exec.sched], schedulers
    struct scheduler_t {};

    ...as before...
}
```

[Editor's note: After `[exec.utils]`, add a new subsection `[exec.envs]` as follows:]

34.12 Queryable utilities `[exec.envs]`

34.12.1 Class template `prop` `[exec.prop]`

```
namespace std::execution {
    template<class QueryTag, class ValueType>
    struct prop {
        QueryTag query_;    // exposition only
        ValueType value_;   // exposition only

        constexpr const ValueType& query(QueryTag) const noexcept {
            return value_;
        }
    };

    template<class QueryTag, class ValueType>
        prop(QueryTag, ValueType)    ->    prop<QueryTag,
            unwrap_reference_t<ValueType>>;
}
```

1. Class template `prop` is for building a queryable object from a query object and a value.
2. *Mandates:* `callable<QueryTag, prop-like<ValueType>>`, where *prop-like* is the following exposition-only class template:

```
template<class ValueType>
struct prop-like { // exposition only
    const ValueType& query(auto) const noexcept;
};
```

3. [Example 1:

```
template<sender Sndr>
sender auto parameterize_work(Sndr sndr) {
    // Make an environment such that get_allocator(env)
    // returns a reference to a copy
    // of my_alloc{}
    auto e = prop(get_allocator, my_alloc{});

    // parameterize the input sender so that it will
    // use our custom execution environment
    return write_env(sndr, e);
}
```

— end example]

4. Specializations of `prop` are not assignable.

34.12.2 Class template `env` [`exec.env`]

```
namespace std::execution {
    template<queryable... Envs>
    struct env {
        Envs0 envs0; // exposition only
        Envs1 envs1; // exposition only
        ...
        Envsn-1 envsn-1; // exposition only

        template<class QueryTag>
        constexpr decltype(auto) query(QueryTag q) const
            noexcept(see below);
    };

    template<class... Envs>
    env(Envs...) -> env<unwrap_reference_t<Envs>...>;
}
```

1. The class template `env` is used to construct a queryable object from several queryable objects. Query invocations on the resulting object are resolved by attempting to query each subobject in lexical order.
2. Specializations of `env` are not assignable.

3. It is unspecified whether `env` supports class template argument deduction ([over.match.class.deduct]) when performing initialization using a parenthesized *expression-list* ([decl.init]).

4. [Example 1:

```
template<sender Sndr>
sender auto parameterize_work(Sndr sndr) {
    // Make an environment such that:
    //   - get_allocator(env) returns a reference to a
    //     copy of my_alloc{}
    //   - get_scheduler(env) returns a reference to a
    //     copy of my_sched{}
    auto e = env{prop(get_allocator, my_alloc{}),
                 prop(get_scheduler, my_sched{})};

    // parameterize the input sender so that it will
    // use our custom execution environment
    return write_env(sndr, e);
}
```

— end example]

34.12.2.1 `env` members [exec.env.members]

```
template<class QueryTag>
constexpr decltype(auto) query(QueryTag q) const noexcept(see
    below);
```

1. Let *has-query* be the following exposition-only concept:

```
template<class Env, class QueryTag>
concept has_query = // exposition
    only
    requires (const Env& env) {
        env.query(QueryTag());
    };
```

2. Let *fe* be the first element of *envs₀*, *envs₁*, ... *envs_{n-1}* such that the expression *fe*.query(*q*) is well-formed.
3. *Requires*: (*has_query*<*Env*s, *QueryTag*> || ...) is true.
4. *Effects*: Equivalent to *fe*.query(*q*).
5. *Remarks*: The expression in the `noexcept` clause is equivalent to `noexcept(fe.query(q))`.

§ 8 Acknowledgements

I would like to thank Lewis Baker for many enlightening conversations about the design of execution environments. The design presented here owes much to Lewis's insights.

§ 9 References

[P3115R0] Corentin Jabot. 2024-02-15. Data Member, Variable and Alias Declarations Can Introduce A Pack.

<https://wg21.link/p3115r0>

[P3149R3] Ian Petersen, Ján Ondrušek; Jessica Wong; Kirk Shoop; Lee Howes; Lucian Radu Teodorescu; 2024-05-22. `async_scope` — Creating scopes for non-sequential concurrency.

<https://wg21.link/p3149r3>

[P3284R0] Eric Niebler. 2024-05-16. `finally`, `write_env`, and `unstoppable` Sender Adaptors.

<https://wg21.link/p3284r0>

1. <https://godbolt.org/z/976b1G45a>↩